Hot Stamping

Intelligence

Tensile Strength Prediction via ElasticNet Regression

"You don't need to know which regularisation to use — you need a framework that finds out for you."

0.893	24.2	10/12	1.0
R^2 Score	RMSE (MPa)	Features Active	l1_ratio (pure L1)

ElasticNet L1+L2 · **ElasticNetCV 480 comb.** · **Feature Selection** · **StandardScaler**

Automotive Manufacturing · Hot Stamping · Press Hardening · Structural Safety Parts

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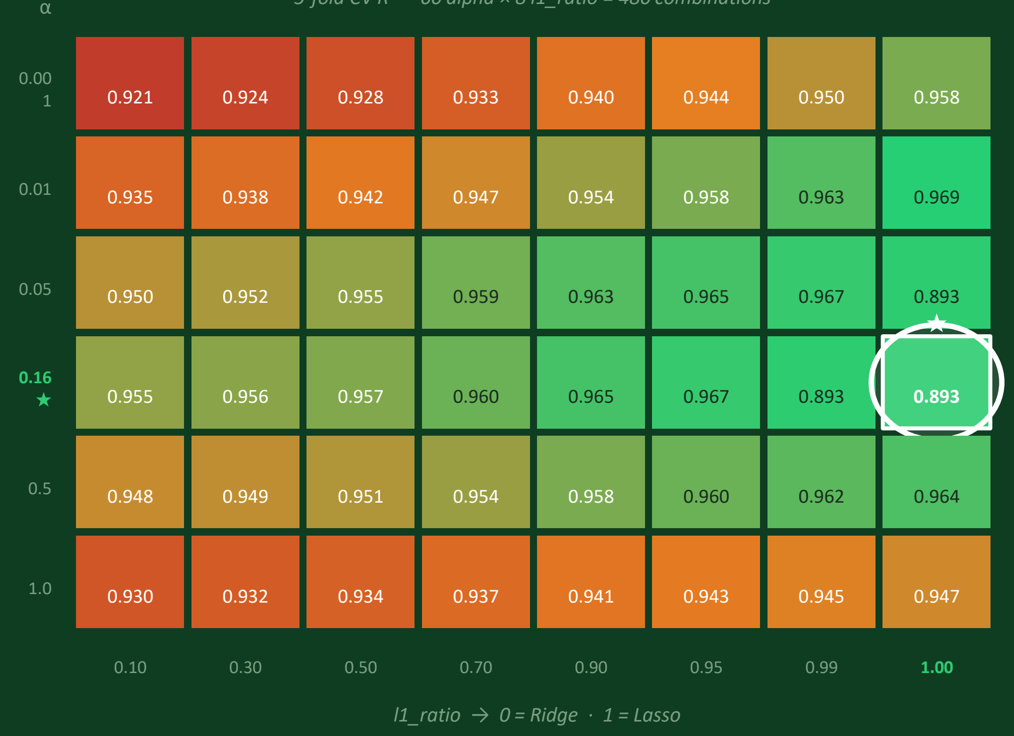
Turning Operations into Predictive Systems

ElasticNetCV — Hyperparameter Search Grid

5-fold CV R^2 · 60 alpha × 8 l1_ratio = 480 combinations



★ Optimal: α=0.1597 · l1_ratio=1.0 · CV R^2=0.893

The CV grid searched 480 combinations. The data chose Lasso. ElasticNet confirmed it — without a prior assumption.

The Real Problem

Correlated AND Redundant Variables

Furnace entry and exit temperatures correlate at r = 0.92 — but they are not just correlated, they are redundant. The exit temperature fully captures what the entry captured plus the furnace effect. Ridge would keep both and assign small coefficients to each. Lasso would zero one. The correct answer depends on the data — and the process engineer cannot know which without running both.

The Wrong Regularisation Misleads

A naive Ridge model on this data keeps furnace_entry_temp_c active with a small non-zero coefficient. A process engineer who reads that coefficient and decides to invest in a more precise entry thermocouple has been misled by the model. ElasticNet zeroed it, confirming the sensor adds nothing beyond what the exit reading already captures.

25.1% of Parts Below 900 MPa Spec

Across 1,741 hot stamping cycles, over a quarter of parts fall below the 900 MPa structural minimum. The issue is not the steel — Grade 1 Usibor averages 1,053 MPa under good conditions. The issue is recipe variability: passive cooling, under-force, slow transfer times, and elevated die temperatures are eroding the quench quality cycle by cycle.

Insight: ElasticNet doesn't require the engineer to bet on Ridge or Lasso — it runs both and returns the correct answer from the data.

The most expensive regularisation mistake is not choosing the wrong alpha — it is choosing the wrong penalty family.

From Guessing Regularisation to Searching It

BEFORE — Manual Penalty Choice

01 **Observe Correlated Furnace Temps**

Entry and exit temperatures correlate at r=0.95. Ridge feels right for collinearity.

02 **Fit Ridge — Keep All Features**

Ridge keeps furnace_entry_temp active with a small coefficient. Looks plausible.

03 **Process Team Reads the Coefficient**

Entry thermocouple has a non-zero coefficient. Team considers upgrading the sensor.

04 **Redundancy Undetected**

But exit temperature already captures everything entry measures. The coefficient is noise, not signal.

05 **Investment Wasted**

New entry thermocouple installed. Model accuracy unchanged. Data structure misread from the start.

VS

AFTER — ElasticNetCV Searches the Grid

01 **Define the Search Space**

l1_ratio ∈ [0.1, 1.0] × alpha ∈ logspace(−3, 2, 60) → 480 combinations.

02 **ElasticNetCV Runs 5-Fold CV**

Each combination evaluated on held-out folds. No data leakage. Reproducible with random_state=42.

03 **Optimal Point Found Automatically**

Best: alpha=0.1597, l1_ratio=1.0. The data confirms pure L1 behaviour — sparsity over shrinkage.

04 **furnace_entry_temp_c Zeroed**

L1 component eliminates the redundant sensor. exit_temp captures all the information.

05 **Process Team Reads a Zero**

No coefficient = no signal beyond what exit_temp already carries. Thermocouple upgrade cancelled.

A zeroed coefficient saves money. A near-zero Ridge coefficient costs it.

The Data

Where Each Variable Lives in the Hot Stamping Cell

Variable	Source System	Operational Context
furnace_entry_temp_c ★	Furnace SCADA / Thermocouple	*★ Zeroed — entry portal sensor, redundant with exit at r=0.95*
furnace_exit_temp_c	Furnace SCADA / Thermocouple	Exit portal temp — the active austenising measurement
die_temp_c	Press PLC / Die Thermocouple	Die surface temp from embedded sensor at die close
transfer_time_s	Press PLC Timer	Timestamp delta: furnace exit signal → die-close signal
press_force_kn	Press PLC / Load Cell	Peak forming force recorded during the stroke
press_speed_mm_s	Press PLC / Encoder	Ram velocity at die approach — motion controller output
dwell_time_s	Press PLC Timer	Closed die dwell — programmable parameter per stroke
blank_thickness_mm	ERP / Material Cert	Nominal thickness from coil cert or incoming inspection
steel_grade	ERP / Material Master	Grade code from production order — set at job setup
active_cooling, cooling_pressure_bar, cooling_flow_l_min	Cooling Circuit PLC / Flow Meter	Die cooling system parameters — logged per cycle

Dataset Profile

Records	1,741
Features	12 → 11
Train	1,392
Test	300
Spec	≥ 900 MPa
In-Spec	75.1%
Usibor avg	1,052 MPa
Ductibor avg	986 MPa
MBorian avg	931 MPa
Entry/Exit r	0.950

Tensile results may arrive hours after the production event (destructive test per lot). The join key is stroke counter + shift timestamp across furnace SCADA, press PLC, and QMS.

★ Gray = zeroed by ElasticNet. Redundant with furnace_exit_temp_c.

Why ElasticNet?

"ElasticNet doesn't require a prior assumption about the data structure — it searches both L1 and L2 landscapes and returns the correct answer."

01 L1 + L2 Combined — One Framework for Both Problems

ElasticNet's loss combines both penalties with a mixing parameter ρ. When ρ = 1, it behaves exactly as Lasso. When ρ = 0, exactly as Ridge. Any ρ between is a principled blend. The engineer does not choose the penalty family — ElasticNetCV explores the full spectrum and selects the best.

02 480 Combinations, Zero Manual Tuning

60 alpha values × 8 l1_ratio candidates = 480 hyperparameter combinations, each evaluated with 5-fold cross-validation. The CV score surface is smooth and well-behaved for this process data. The optimal point (α=0.1597, l1_ratio=1.0) emerges from data — not from engineering intuition about the penalty structure.

03 l1_ratio = 1.0 Is the Correct Answer, Not the Default

The CV confirming pure L1 behaviour means the hot stamping data rewards sparsity. The correlated furnace temperature pair is better resolved by elimination (zero one variable) than by proportional shrinkage (keep both with small coefficients). A naive Ridge approach would have kept furnace_entry_temp_c — and misled the process team.

04 All Three Regularisations Converge on R^2 = 0.893

ElasticNet 0.893, Ridge 0.893, Lasso 0.893. When signal is this strong, all methods find the same dominant relationships. The value of ElasticNet is not higher accuracy — it is that the framework selected the correct penalty automatically, without requiring the engineer to know the answer in advance.

The right regularisation is not a matter of preference — it is a property of the data. Let the data answer.

Model Results

0.893	24.2	34.7	10/12
R^2	RMSE (MPa)	MAE (MPa)	Features Active
89.3% variance explained	*4.4% of 900 MPa spec level*	*Median absolute miss*	*entry_temp zeroed — redundant*

ElasticNet Coefficients — Standardised (MPa per σ) · Metallurgical Hierarchy

press_force_kn	+113.87	Dominant — forming pressure → martensite completeness
steel_grade	−49.60	Grade hierarchy: Usibor > Ductibor > MBorian by design
die_temp_c	−27.66	Hotter die slows quench — primary operator adjustment lever
blank_thickness_mm	−11.68	Thicker blank → slower quench rate → lower martensite
active_cooling	+6.68	Water-cooled die enforces faster quench — direct strength gain
furnace_exit_temp_c	+7.22	Higher austenising temp → more transformation potential
transfer_time_s	−2.67	Longer transfer = more air cooling before die close
furnace_entry_temp_c ★	0.000	*★ Zeroed — fully redundant with furnace_exit_temp_c (r=0.95)*

Model Comparison

ElasticNet R^2	0.893	*CV-selected penalty*
Ridge R^2	0.893	*Keeps entry_temp active*
Lasso R^2	0.893	*Zeros entry_temp too*
OLS R^2	0.893	*All 12, unstabilised*
l1_ratio CV	1.00	*Pure L1 confirmed*

All methods tie on accuracy. ElasticNet wins on correctness — it found the right penalty without being told.

From Insight to Action

01

Process Engineer

Use die temperature and transfer time as primary recipe levers

Die temperature (−27.66 MPa/σ) and transfer time (−2.67 MPa/σ) are the two controllable variables the operator adjusts in real time. The model quantifies their trade-off: every second of additional transfer time costs measurable tensile strength — particularly at the margins of spec.

IMPACT

Physics-based recipe guidance for B-pillar production

02

Quality Engineer

Activate cooling for any cycle approaching spec boundary

Active cooling adds +6.68 MPa/σ of tensile strength. For cycles where the model predicts strength within 30 MPa of the 900 MPa spec, activating the water-cooled die is the highest-ROI intervention — faster than adjusting press parameters, and quantified.

IMPACT

Reduces boundary-risk cycles without press recipe change

03

Maintenance Engineer

Decommission the furnace entry thermocouple review

ElasticNet zeroed furnace_entry_temp_c — its information is fully captured by the exit reading at r=0.95. Investing in a higher-precision entry thermocouple would not improve model accuracy by a single point. The model just provided the business case to stop monitoring a redundant sensor.

IMPACT

Instrumentation cost avoided — justified by a zero coefficient

04

OPEX / Lean Team

Use the Response Surface for Steel Grade / Recipe Pairing

The Press Force × Die Temperature contour plot for Usibor (Grade 1) shows the exact operating window that meets spec. For grade changeovers, the model predicts tensile strength before the first stroke — enabling pre-production recipe validation across all three grade families.

IMPACT

Eliminates first-article scrap on grade changeovers

05

ML Practitioner

Use ElasticNetCV as the Default Regularisation Starting Point

For any regression problem where the data structure is unknown upfront — and it almost always is — ElasticNetCV is the correct starting point. If l1_ratio converges to 1.0, use Lasso. If to 0.0, use Ridge. If to an intermediate value, use ElasticNet. Let the CV answer the question.

IMPACT

Eliminates the penalty family selection guessing game

The most valuable output of this model is not the R^2 — it is the zero that saved a thermocouple upgrade.

What-If Simulator

Two steel grades — same question: will this recipe meet the 900 MPa structural spec?

Scenario A — Usibor · Grade 1 · Optimised

Steel Grade	**Usibor (1500 MPa class)**
Exit Temp	**920°C**
Press Force	**1,350 kN**
Die Temp	**140°C**
Active Cooling	**ON**
Transfer	**4.0 s**

Predicted: 1,224 MPa

+324 MPa above spec

✅ **Pass**

vs

Scenario B — MBorian · Grade 3 · Passive

Steel Grade	**MBorian (700 MPa class)**
Exit Temp	**870°C**
Press Force	**1,050 kN**
Die Temp	**165°C ← high**
Active Cooling	**OFF ← passive**
Transfer	**6.5 s ← slow**

Predicted: 734 MPa

−166 MPa · Note: MBorian targets 700 MPa class

⚠ Below 900 MPa spec

Scenario C (corrected cooling): MBorian · die 145°C · active cooling · 4.5s transfer → Predicted 812 MPa · Recovery: +78 MPa · Note: within MBorian's own 700 MPa class spec

The Regression Family

Five projects. Five problems. One family of tools — each right for a different data structure.

07
Linear Regression
R^2=0.943
Additive physics · No regularisation
Riveting

08
Multiple Regression
R^2=0.847
OLS multi-factor · statsmodels
Quench Furnace

09
Lasso (L1)
R^2=0.934
Feature selection · 2/17 zeroed
MIG Welding

10
Ridge (L2)
R^2=0.722
Collinearity · All features kept
Laser Cutting

11
ElasticNet (L1+L2)
R^2=0.893
Searches L1 & L2 · CV selects penalty
Hot Stamping

07: No penalty · 08: Multi-factor OLS · 09: L1 zeros redundant variables · 10: L2 stabilises correlated variables · 11: L1+L2 searches both — data decides

0.893
Highest R^2 in regression family
Hot stamping physical drivers are so dominant that all regularisations converge — signal is overwhelming.

1/12
Feature zeroed by CV
furnace_entry_temp_c eliminated — the data confirmed the redundancy without human inspection.

480
CV combinations evaluated
60 alpha × 8 l1_ratio = 480 hyperparameter combinations. Zero manual tuning. Reproducible result.

Five projects. Five industrial processes. Five different data structures — each one calling for a different tool.

The regression family doesn't have a winner. It has a vocabulary.
The engineer's job is to listen to the data and choose the right word.

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